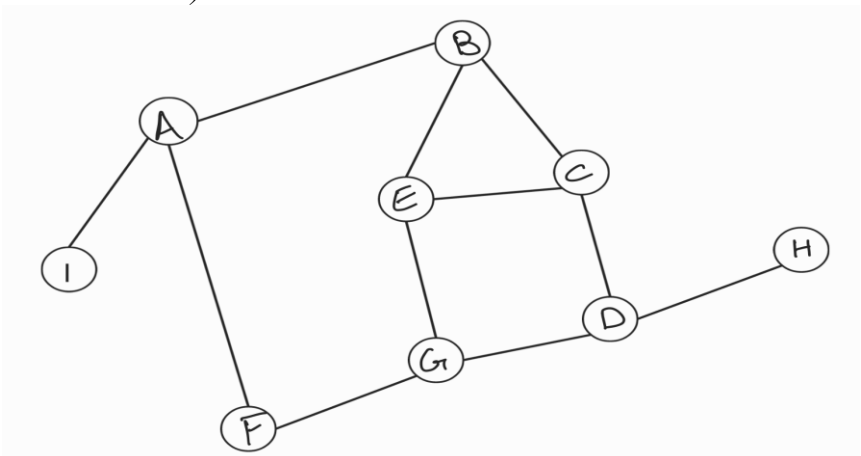
Tutorial Quiz
Set A

Instruction: **Max. Marks: 10**
Write in clear handwriting.

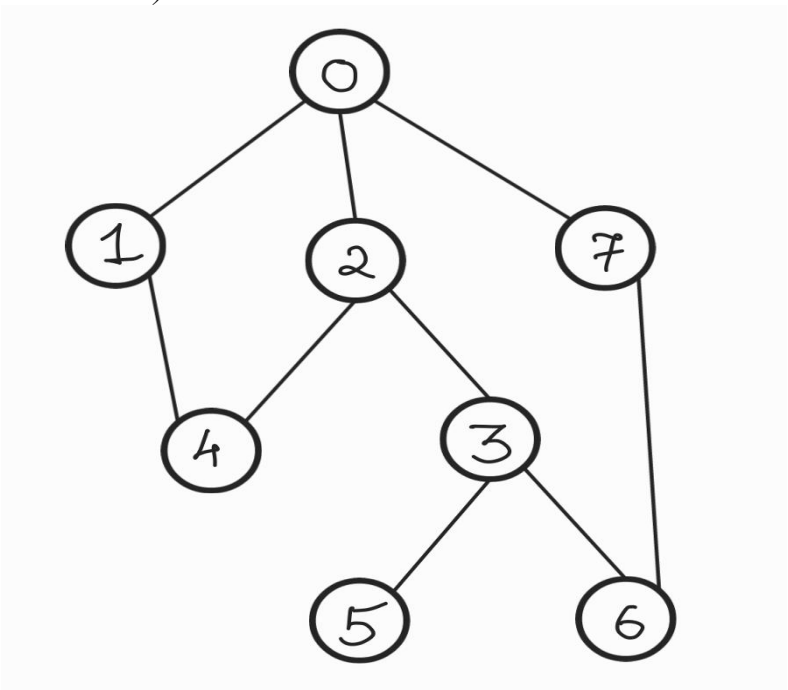
Q1. **(A)** Perform a **Breadth-First Search (BFS)** starting from node **A**, giving preference to vertices with **lower alphabetical order** when multiple choices are available.

- Show the **intermediate states of the queue** and illustrate the **step-by-step traversal of the graph**.
- Compute the **shortest path distances** from **A** to vertices **E**, **G**, and **H**.
  (2 + 0.5 = 2.5 marks)

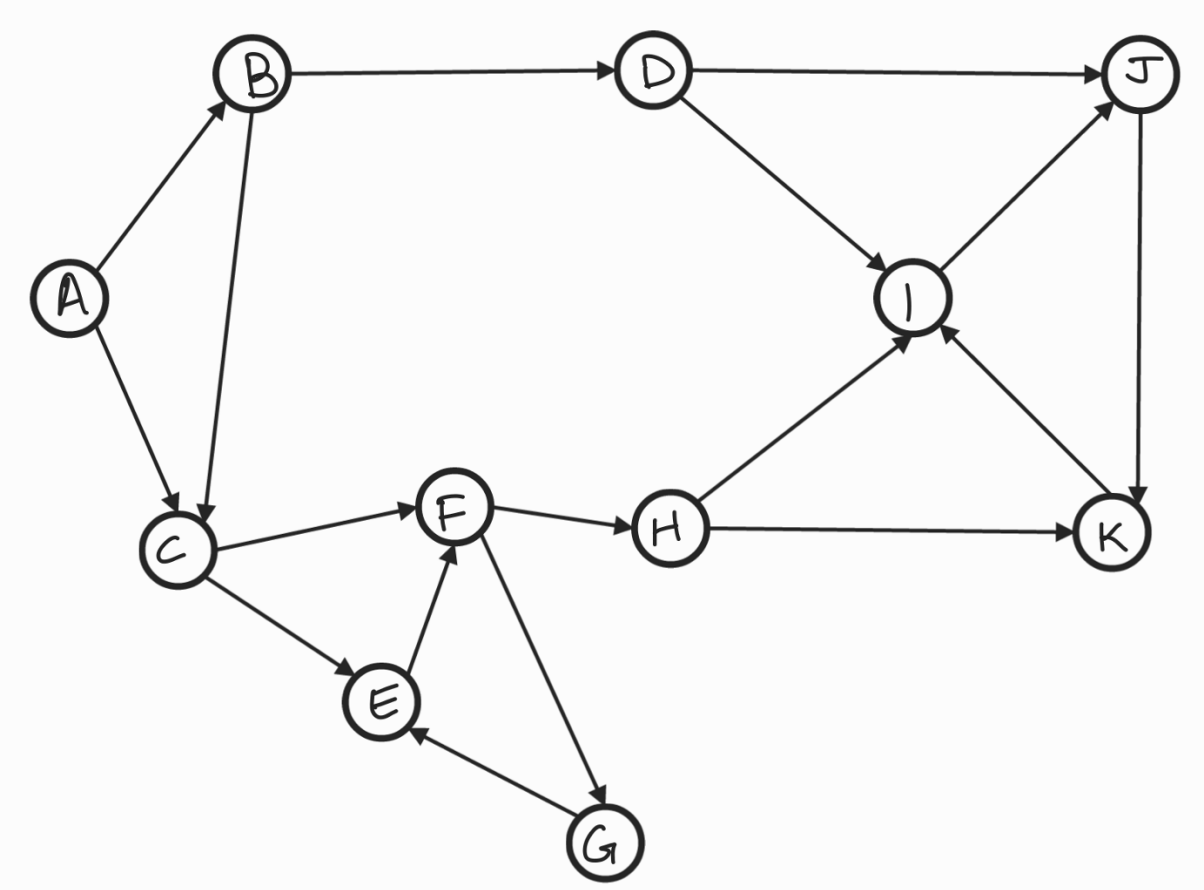


(B) Perform a **Breadth-First Search (BFS)** starting from node **0**, giving preference to vertices with **smaller numeric labels** when multiple choices are available.

- Show the **intermediate states of the queue** and illustrate the **step-by-step traversal of the graph**.
- Compute the **shortest path distances** from **0** to vertices **5** and **6**.
  (2 + 0.5 = 2.5 marks)



Q2. Perform a **Depth-First Search (DFS)** on the following directed graph G, starting from node A.
Give preference to visiting lower-numbered vertices before higher-numbered vertices whenever multiple choices are available.
(a) Write down the **discovery and finishing times** of all vertices.
(1 mark)
(b) Draw the **DFS forest**.
(1.5 marks)
(c) Classify all edges of the graph as **Tree, Back, Forward, or Cross edges**.
(2 marks)
(d) Determine whether the graph contains a **cycle**. Answer Yes or No, and provide a brief reason.
(0.5 mark)

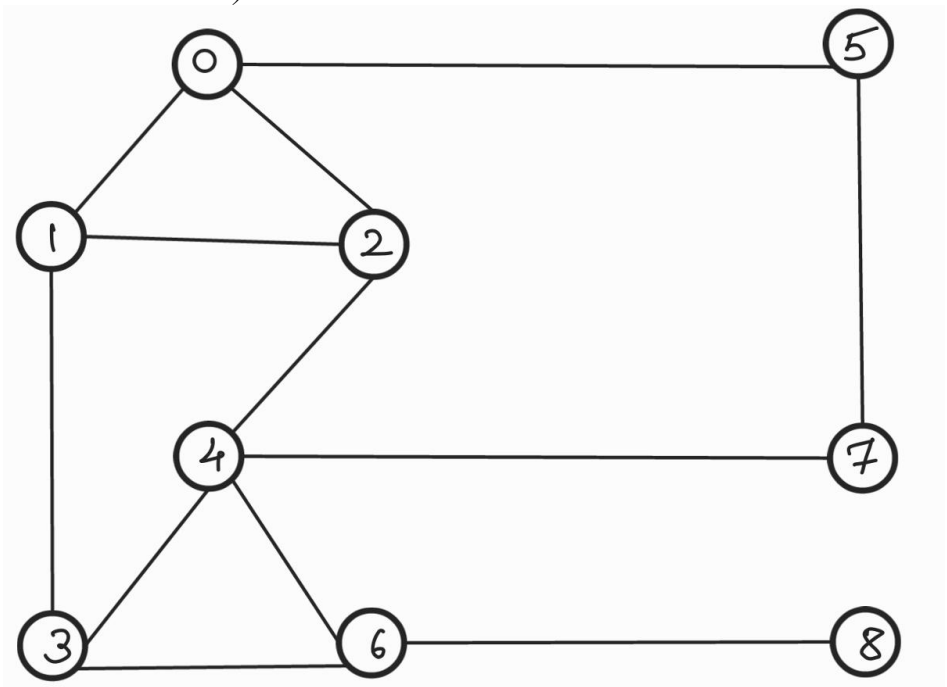


Tutorial Quiz
Set B

Instructions: **Max. Marks: 10**
Write in clear handwriting.

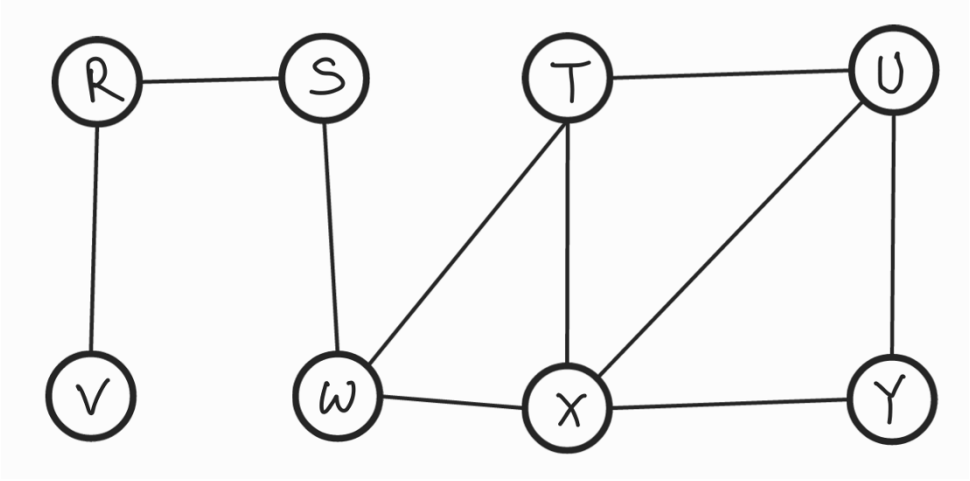
Q1. **(A)** Perform a **Breadth-First Search (BFS)** starting from node **0**, giving preference to vertices with **smaller numeric labels** when multiple choices are available.

- Show the **intermediate states of the queue** and illustrate the **step-by-step traversal of the graph**.
- Calculate the **shortest path distances** from **0** to vertices **4**, **6**, and **7**.
  (2 + 0.5 = 2.5 marks)

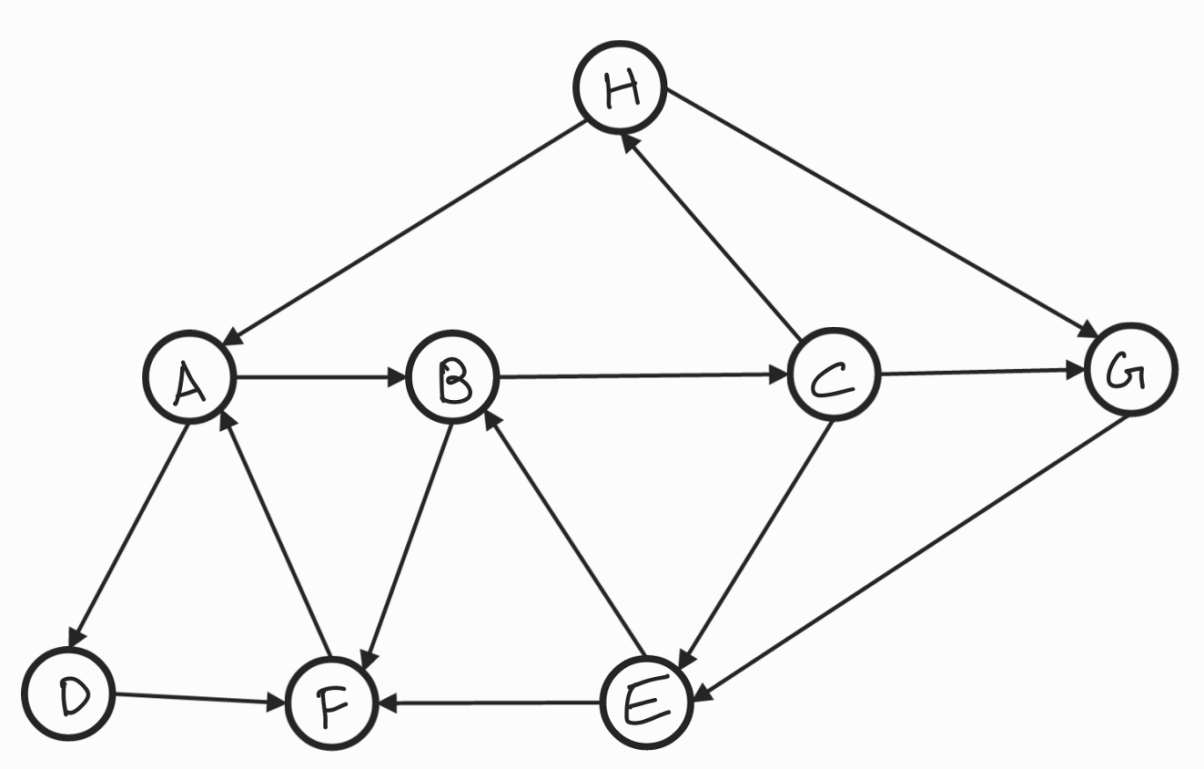


(B) Perform a **Breadth-First Search (BFS)** starting from node **s**, giving preference to vertices with **lower alphabetical order** when multiple choices are available.

- Show the **intermediate states of the queue** and illustrate the **step-by-step traversal of the graph**.
- Calculate the **shortest path distances** from **s** to vertices **u** and **y**.
  (2 + 0.5 = 2.5 marks)



Q2. Perform a **Depth-First Search (DFS)** on the following directed graph G, starting from node A.
Give preference to visiting lower-numbered vertices before higher-numbered vertices whenever multiple choices are available.
(a) Write down the **discovery and finishing times** of all vertices.
(1 mark)
(b) Draw the **DFS forest**.
(1.5 marks)
(c) Classify all edges of the graph as **Tree, Back, Forward, or Cross edges**.
(2 marks)
(d) Determine whether the graph contains a **cycle**. Answer Yes or No, and provide a brief reason.
(0.5 mark)

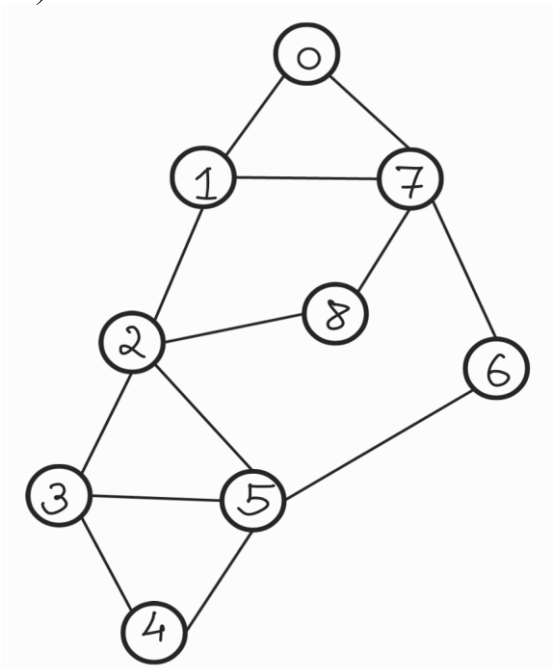


Tutorial Quiz
Set C

Instructions: **Max. Marks: 10**
Write in clear handwriting.

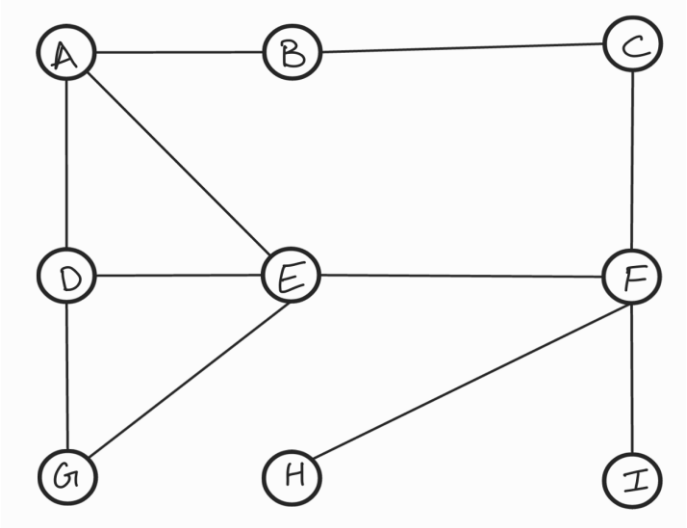
Q1. **(A)** Perform a **Breadth-First Search (BFS)** starting from node **0**, giving preference to vertices with **smaller numeric labels** when multiple choices are available.

- Show the **intermediate states of the queue** and illustrate the **step-by-step traversal of the graph**.
- Calculate the **shortest path distances** from **0** to vertices **2**, **4**, and **5**.
  (2 + 0.5 = 2.5 marks)



(B) Perform a **Breadth-First Search (BFS)** starting from node **A**, giving preference to vertices with **lower alphabetical order** when multiple choices are available.

- Show the **intermediate states of the queue** and illustrate the **step-by-step traversal of the graph**.
- Calculate the **shortest path distances** from **A** to vertices **E** and **G**.
  (2 + 0.5 = 2.5 marks)



Q2. Perform a **Depth-First Search (DFS)** on the following directed graph G, starting from node A.
Give preference to visiting lower-numbered vertices before higher-numbered vertices whenever multiple choices are available.
(a) Write down the **discovery and finishing times** of all vertices.
(1 mark)
(b) Draw the **DFS forest**.
(1.5 marks)
(c) Classify all edges of the graph as **Tree, Back, Forward, or Cross edges**.
(2 marks)
(d) Determine whether the graph contains a **cycle**. Answer Yes or No, and provide a brief reason.
(0.5 mark)

